

Sample Paper 5 — Answer Key

Total: 60 marks (45×1 mark + 5×3 marks), 60 minutes.

Section 1 — Logical Reasoning

1. **D) Peacock** — Lion, Tiger, and Leopard are big cats; Peacock is a bird.
2. **B) Speed** — A thermometer measures temperature; a speedometer measures speed.
3. **A) GHVN** — The rule is "+3 letters": $D \rightarrow G$, $E \rightarrow H$, $S \rightarrow V$, $K \rightarrow N$.
4. **C) 176** — Each term doubles: 11, 22, 44, 88, 176.
5. **B) 17th** — Position from the right = $35 - 19 + 1 = 17$.
6. **C) North** — Two 90° anticlockwise turns from South = 180° turn = North.
7. **A) 2, 1, 4, 3** — A plant's growth: Seed $\rightarrow$ Seedling $\rightarrow$ Sapling $\rightarrow$ Flowering Plant.
8. **C) 53** — 33, 44, and 66 are multiples of 11; 53 is not.
9. **B) Friday** — 45 days = 6 full weeks (42 days, same day: Tuesday) + 3 more days $\rightarrow$ Wed, Thu, Fri.
10. **B) Cousin** — Suresh's father's only sister is his paternal aunt; her daughter is his cousin.
11. **C) 127** — The pattern is "double and add 1": $7 \times 2 + 1 = 15$, $15 \times 2 + 1 = 31$, $31 \times 2 + 1 = 63$, $63 \times 2 + 1 = 127$.
12. **A) EYPBCL** — The rule is "-2 letters" (A wraps to Y): $G \rightarrow E$, $A \rightarrow Y$, $R \rightarrow P$, $D \rightarrow B$, $E \rightarrow C$, $N \rightarrow L$.
13. **C) Rectangle** — Pyramid, Cone, and Cylinder are 3-D solids; a Rectangle is a flat (2-D) shape.
14. **C) C** — Order: $C > A > B$, and D is last — so C finished first.
15. **C) 38** — The gaps increase by 2 each time (+3,+5,+7,+9,+11): 3,6,11,18,27,38.

Section 2 — Mathematical Reasoning

16. **A) 74,134** — $46,789 + 27,345 = 74,134$.
17. **A) 40,528** — $80,006 - 39,478 = 40,528$.
18. **A) 34,068** — $5,678 \times 6 = 34,068$.
19. **A) 9,639** — $57,834 \div 6 = 9,639$.
20. **C) 18** — HCF of 54 (2×3^3) and 72 ($2^3 \times 3^2$) is $2 \times 3^2 = 18$.
21. **A) 42** — Multiples of 14: 14,28,42... Multiples of 21: 21,42... First common one = 42.
22. **A) 4/18** — Multiply top and bottom of $2/9$ by 2.
23. **A) 10/11** — Same denominator: $6+3+1=10$, so $10/11$.
24. **A) 16/25** — $0.64 = 64/100$, which simplifies ($\div 4$) to $16/25$.
25. **C) 121** — 55% of 220 = $0.55 \times 220 = 121$.

26. **C) 52%** — $13/25 = 0.52 = 52\%$.
27. **C) 44** — 1 part = $16 \div 4 = 4$; larger = 11 parts = 44.
28. **B) 20%** — Loss = $850 - 680 = ₹170$; loss % = $170/850 \times 100 = 20\%$.
29. **C) ₹1,800** — SI = $(9000 \times 5 \times 4) \div 100 = ₹1,800$.
30. **A) 7,008** — 7 kg = 7,000 g; + 8 g = 7,008 g.
31. **C) 195** — $3.25 \times 60 = 195$ minutes.
32. **B) 130 cm** — Perimeter = $2 \times (42 + 23) = 130$ cm.
33. **B) 289 sq cm** — Area = $17 \times 17 = 289$ sq cm.
34. **C) 4** — A square has 4 lines of symmetry.
35. **B) 3** — Three-quarters of $360^\circ = 270^\circ$; $270 \div 90 = 3$.

Section 3 — Everyday Mathematics

36. **C) 54,400** — $3,400 \times 16 = 54,400$.
37. **B) ₹2,275** — $6400 - 1950 - 2175 = 2,275$.
38. **C) 450 L** — $4/9$ of 810 = 360 L used; $810 - 360 = 450$ L left.
39. **C) ₹45** — $1,215 \div 27 = 45$.
40. **B) 323** — $17 \times 19 = 323$.
41. **C) ₹2,100** — SI = $(6000 \times 7 \times 5) \div 100 = ₹2,100$.
42. **B) 18%** — Profit = $3304 - 2800 = ₹504$; profit % = $504/2800 \times 100 = 18\%$.
43. **C) 84 km** — 1 h 20 min = $4/3$ hours; $63 \times 4/3 = 84$ km.
44. **B) 6 cups** — $3/10 \times 20 = 6$ cups.
45. **C) 140** — 40% of 350 = $0.40 \times 350 = 140$.

Section 4 — Achievers Section (3 marks each)

46. **C) 300** — If the original number is x : $1.45x = 435 \rightarrow x = 300$.
47. **C) 30** — Five consecutive numbers $n-2$ through $n+2$ sum to $5n = 150 \rightarrow n = 30$ (the middle number).
48. **B) ₹7,000** — SI = $(P \times 3 \times 8) \div 100 = 0.24 \times P = 1,680 \rightarrow P = 1,680 \div 0.24 = ₹7,000$.
49. **C) ₹3,000** — After snacks, $5/6$ of the money remains. After books ($2/5$ of that spent), $3/5$ of $5/6 = 1/2$ of the original remains. So $1/2$ of the total = ₹1,500 $\rightarrow$ total = ₹3,000.
50. **C) 20** — Let the numbers be $3x$ and $4x$. After decreasing each by 5: $(3x-5)/(4x-5) = 2/3 \rightarrow 3(3x-5) = 2(4x-5) \rightarrow 9x-15 = 8x-10 \rightarrow x = 5$. Larger number = $4 \times 5 = 20$.
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Scoring guide (out of 60): 42+ is Level-2 cutoff range — if you're here, you're genuinely ready for the real exam. 30–41 is solid Level-1 performance. Below 30 — one more review pass over the weak topics before exam day, then attempt Sample Paper 6 (the harder, Level-2-style paper) if time allows.